Michael King

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SUMMARY

Mechanical engineer with five years at a robotics manufacturer that ships computer-vision-powered robotic cells. Holds the Operations sign-off on product software releases and built the system that tracks what is engineered, built, and shipped. Turns field failures into tests, gates, and repeatable environments. Independently builds evaluation harnesses and automated checks for AI systems, and publishes the results, misses included.

PROFESSIONAL EXPERIENCE

Plus One Robotics (computer-vision and AI robotics for parcel handling) · San Antonio, TX — Mar 2021 - Present

Operations Systems Engineer II — Mar 2026 - Present

- Specified and directed a desk-side integration rig (two IPCs, an HMI, a PLC, two depth cameras, and five Pi CM4 cameras networked as a complete cell) after finding that releases shipped to the field with no integration test; every release is now checked against real device handshakes before it ships.
- Serve as the engineering release gate for every product's drawings, BOMs, and electrical schedules, each linked back to its part-number allocation; hold Operations sign-off on product software releases, shipped as orchestrated Docker images, with linked evidence required for every requirement.
- Sole developer of InductOne Tools, a Frappe/ERPNext application that runs the robot product line from engineering configuration through supplier handoff to field service: about 33,500 lines, 50 custom DocTypes, and 24 migration patches in 5.5 months. It carries roughly 1,100 items, 277 BOMs, 93 engineering signoffs, and 90 allocated part numbers for 30 users.
- Built configuration status accounting after observing where early adoption broke down; ERPNext is now the source of truth for product releases, with a defined release gate and nothing left in limbo.
- Head Ops assembly of the Individual Cup Control gripper; turned an unsupported service process into an in-field maintenance test and compensator replacement guide, shifting routine service to the customer with Product and Engineering approval.
- Designed the external-builder workflow: row-level permissions limit each supplier to its own builds, releases pass readiness gates, and acceptance creates a locked as-built record in a single transaction.
- Replaced a manual builder-package process estimated at 200 hours per machine with a generator that also produces RFQ packages; modeled value of $43K to $56K per year in recovered engineering time at the current build rate.
- Integrated ERPNext with GitLab, the production imaging system, and device log imports from IPC, HMI, and camera units.
- Build every new production IPC, run inventory including physical counts for the company-wide audit, and manage freight and parcel shipping accounts.

Operations Systems Engineer — Nov 2025 - Mar 2026

- Wrote the company's configuration status accounting framework and serialization strategy (ISO 10007, EIA-649C); designed the BOM architecture and variant logic for high-mix, low-volume manufacturing.
- Gathered ERP requirements from engineering, procurement, software, operations, and leadership; vetted platforms against a written requirements table and drove adoption across every department to 30 active users.

Production Engineer — Dec 2022 - Nov 2025

- Owned production systems between engineering and manufacturing; built commissioning tooling and runbooks for robot deployments in the US and EU.
- Set up and ran OS imaging for Ubuntu production machines, including LAN-based deployment and remote troubleshooting of live units.
- Built shop-floor tools, including an OpenCV station that verifies QA labels by OCR, packaged as a Windows executable for operators.

Systems Integration Engineering Intern — Mar 2021 - Dec 2022

- Wrote Python GUI and CLI tools that error-proofed documentation and manufacturing QA records; supported hardware deployment and converted to full-time.

INDEPENDENT PROJECTS

- **OnScript** - Daily public-data measurement pipeline that publishes a static site where every claim links to its sources. LLM classification is held behind a published gold-set harness and a classifier floor audit. Runs unattended on GitHub Actions with fail-closed checks, a published methodology, and a corrections log. onscript.news · github.com/mlawsonking/onscript
- **Agent Guards** - Deterministic checks for AI agent workflows that produce reproducible evidence for known patterns, shipped as MCP servers, an API, a Claude plugin, and a GitHub Action. The published corpus catches 32 of 40 cases with 0 false positives on 11 ordinary ones; misses, including a reworded prompt injection, are published too. github.com/mlawsonking/MCP
- **The Exhaust** - Automated collectors over public NHTSA, CMS, WARN, and CPSC data with preregistered evaluations; the first two forecasts missed their targets, and the writeups are published as they happened. Built through a multi-agent process with an adversarial review gate at every acceptance, which found 9, 19, and 21 real defects across three passes. github.com/mlawsonking/theexhaust
- **ForgeSense** - Built an edge sensing and analysis platform spanning embedded hardware, signal processing, an edge service, and a web dashboard.

SKILLS

AI evaluation: gold-set harnesses, automated judges, preserved-miss test corpora, preregistered evaluations, multi-agent orchestration (Claude, Codex, MCP)

Technical: Python, Frappe/ERPNext, REST APIs and systems integration, SQL, Docker-based deployment, Bash/Linux, Git/GitLab, CI/CD (GitLab CI, GitHub Actions)

Engineering and manufacturing: hardware-in-the-loop release testing, configuration management (CSA, ECR/ECO/ECN), configurable BOMs, SolidWorks/CAD, manufacturing QA

EDUCATION

B.S. Mechanical Engineering, The University of Texas at San Antonio, Dec 2022. *Summa cum laude*, 4.0 GPA.

B.S. Psychology, Texas A&M University, 2010.